

Task 4: Classification Models, Evaluation Metrics & Handling Imbalanced Data

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1. Metric Selection

A binary classification system was built on the Breast Cancer Wisconsin dataset (569 samples, 30 features) to predict Malignant (0) vs Benign (1) tumors. The dataset is moderately imbalanced (~63% Benign, ~37% Malignant), so accuracy alone was judged insufficient: a model predicting "Benign" for every sample would still score ~63% accuracy while being clinically useless. Precision, Recall, and F1-score were used instead, since they separate the two error types. In a medical context, missing a real malignant tumor is far more costly than a false alarm, so Recall on the Malignant class and the overall F1-score were prioritized over raw accuracy when judging model quality.

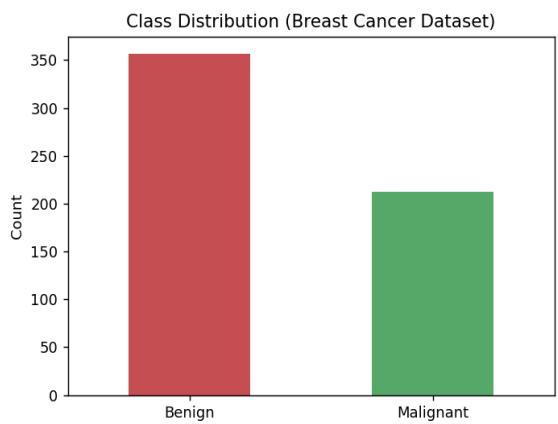


Figure 1: Class distribution — a moderate imbalance (63% Benign vs 37% Malignant).

2. Confusion Matrix & ROC Analysis

The baseline Logistic Regression model was trained on scaled features with a stratified 80/20 train-test split (preserving class proportions). It achieved a near-perfect confusion matrix (41 True Negatives, 71 True Positives, only 1 False Positive and 1 False Negative out of 114 test samples), and an AUC of 0.995, indicating the model separates the two classes almost perfectly across all decision thresholds.

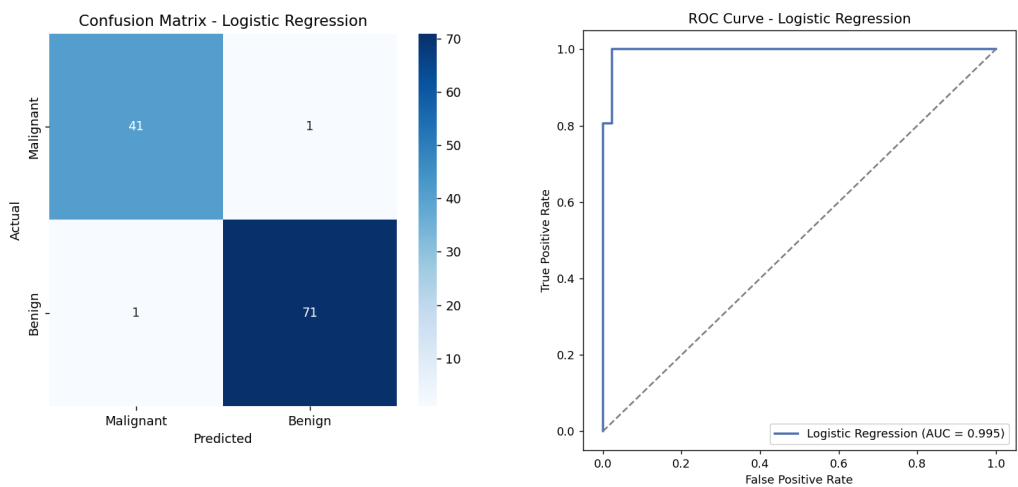


Figure 2 & 3: Confusion matrix and ROC curve for the baseline Logistic Regression model.

3. Imbalance Handling

Class weighting (`class_weight='balanced'`) was tested as an imbalance-handling technique, which penalizes errors on the minority class more heavily during training. On this dataset, the technique was found unnecessary: the baseline model already achieved near-perfect recall (0.986) on the Malignant class because the imbalance is mild (63/37) and the classes are well-separated in feature space. The class-weighted model actually scored slightly lower on accuracy, precision, and F1 than the baseline — a reminder that imbalance-handling techniques should be applied only when they measurably help, and matter far more on severely imbalanced datasets (e.g. 99/1 fraud detection) than on datasets like this one.

4. Model Comparison & Final Decision

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression (baseline)	0.9825	0.9861	0.9861	0.9861
Logistic Regression (balanced)	0.9561	0.9855	0.9444	0.9645
Decision Tree	0.9123	0.9559	0.9028	0.9286

(Baseline Logistic Regression row highlighted — best on every metric.)

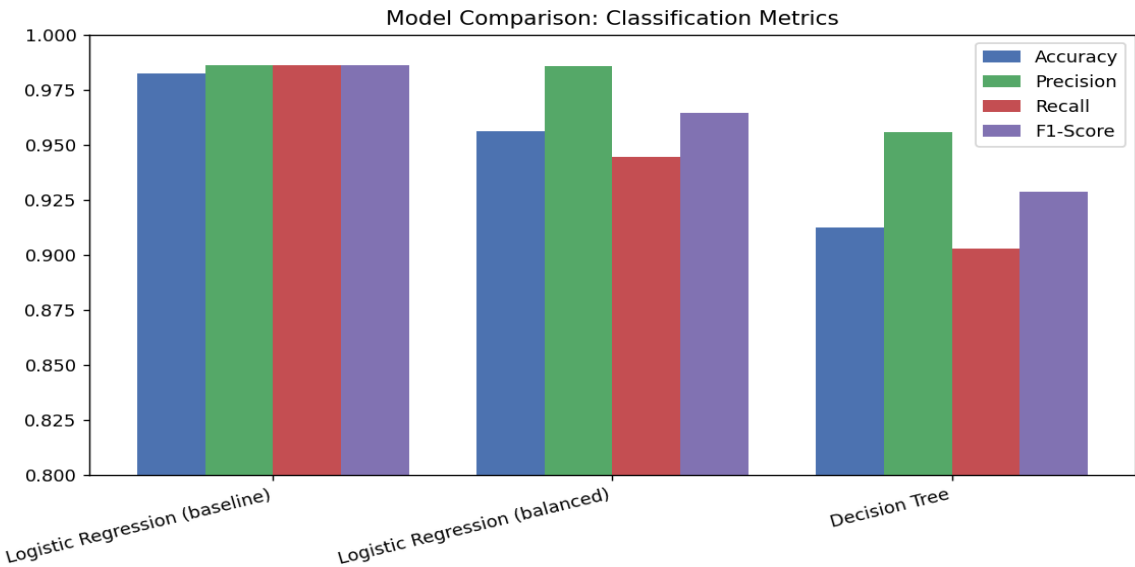


Figure 4: Side-by-side comparison of all three models across four metrics.

Final Model Decision

The **baseline Logistic Regression** model is selected as the final model. It achieved the best scores across every metric (Accuracy 0.983, Precision 0.986, Recall 0.986, F1 0.986) and the highest ROC-AUC (0.995), while remaining simple, fast to train, and easy to interpret via its coefficients. The Decision Tree, trained without depth constraints or tuning, overfit more and generalized worse on this dataset (Accuracy 0.912). Logistic Regression's smoother, well-calibrated decision boundary suits this relatively clean, near-linearly-separable dataset better than a single unconstrained tree.

5. Deliverables

AI_ML_Task4_Classification_Evaluation.ipynb (confusion matrix, classification report, ROC curve, class imbalance handling, model comparison), this PDF report, and the saved models (logistic_model.joblib, logistic_balanced_model.joblib, scaler.joblib).